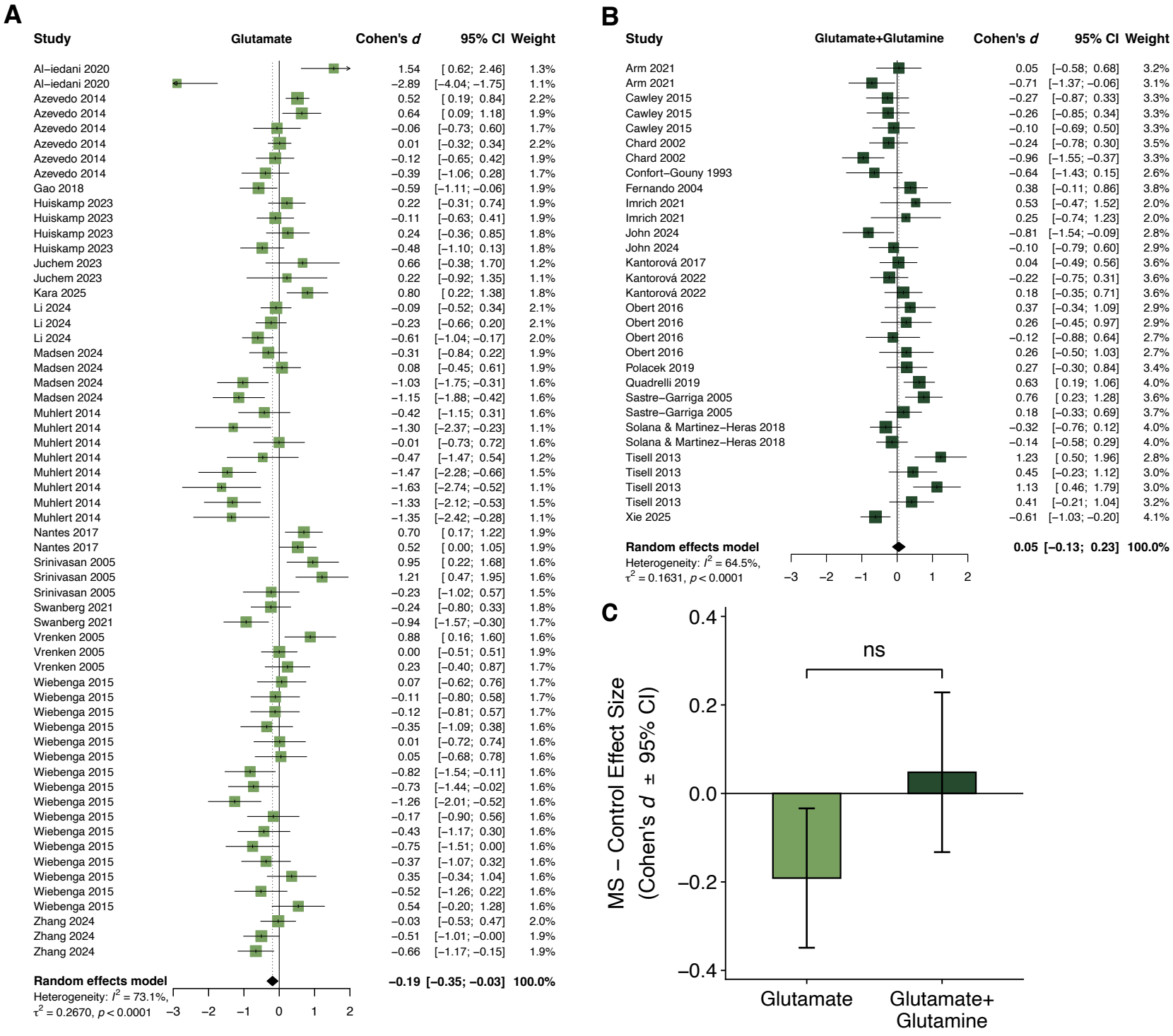

A
Study
Glutamate
Cohen's d
95% CI
Weight
Al–iedani 2020 1.54 [0.62; 2.46] 1.3%
Al–iedani 2020 −2.89 [−4.04; −1.75] 1.1%
Azevedo 2014 0.52 [0.19; 0.84] 2.2%
Azevedo 2014 0.64 [0.09; 1.18] 1.9%
Azevedo 2014 −0.06 [−0.73; 0.60] 1.7%
Azevedo 2014 0.01 [−0.32; 0.34] 2.2%
Azevedo 2014 −0.12 [−0.65; 0.42] 1.9%
Azevedo 2014 −0.39 [−1.06; 0.28] 1.7%
Gao 2018 −0.59 [−1.11; −0.06] 1.9%
Huiskamp 2023 0.22 [−0.31; 0.74] 1.9%
Huiskamp 2023 −0.11 [−0.63; 0.41] 1.9%
Huiskamp 2023 0.24 [−0.36; 0.85] 1.8%
Huiskamp 2023 −0.48 [−1.10; 0.13] 1.8%
Juchem 2023 0.66 [−0.38; 1.70] 1.2%
Juchem 2023 0.22 [−0.92; 1.35] 1.1%
Kara 2025 0.80 [0.22; 1.38] 1.8%
Li 2024 −0.09 [−0.52; 0.34] 2.1%
Li 2024 −0.23 [−0.66; 0.20] 2.1%
Li 2024 −0.61 [−1.04; −0.17] 2.0%
Madsen 2024 −0.31 [−0.84; 0.22] 1.9%
Madsen 2024 0.08 [−0.45; 0.61] 1.9%
Madsen 2024 −1.03 [−1.75; −0.31] 1.6%
Madsen 2024 −1.15 [−1.88; −0.42] 1.6%
Muhlert 2014 −0.42 [−1.15; 0.31] 1.6%
Muhlert 2014 −1.30 [−2.37; −0.23] 1.1%
Muhlert 2014 −0.01 [−0.73; 0.72] 1.6%
Muhlert 2014 −0.47 [−1.47; 0.54] 1.2%
Muhlert 2014 −1.47 [−2.28; −0.66] 1.5%
Muhlert 2014 −1.63 [−2.74; −0.52] 1.1%
Muhlert 2014 −1.33 [−2.12; −0.53] 1.5%
Muhlert 2014 −1.35 [−2.42; −0.28] 1.1%
Nantes 2017 0.70 [0.17; 1.22] 1.9%
Nantes 2017 0.52 [0.00; 1.05] 1.9%
Srinivasan 2005 0.95 [0.22; 1.68] 1.6%
Srinivasan 2005 1.21 [0.47; 1.95] 1.6%
Srinivasan 2005 −0.23 [−1.02; 0.57] 1.5%
Swanberg 2021 −0.24 [−0.80; 0.33] 1.8%
Swanberg 2021 −0.94 [−1.57; −0.30] 1.7%
Vrenken 2005 0.88 [0.16; 1.60] 1.6%
Vrenken 2005 0.00 [−0.51; 0.51] 1.9%
Vrenken 2005 0.23 [−0.40; 0.87] 1.7%
Wiebenga 2015 0.07 [−0.62; 0.76] 1.7%
Wiebenga 2015 −0.11 [−0.80; 0.58] 1.7%
Wiebenga 2015 −0.12 [−0.81; 0.57] 1.7%
Wiebenga 2015 −0.35 [−1.09; 0.38] 1.6%
Wiebenga 2015 0.01 [−0.72; 0.74] 1.6%
Wiebenga 2015 0.05 [−0.68; 0.78] 1.6%
Wiebenga 2015 −0.82 [−1.54; −0.11] 1.6%
Wiebenga 2015 −0.73 [−1.44; −0.02] 1.6%
Wiebenga 2015 −1.26 [−2.01; −0.52] 1.6%
Wiebenga 2015 −0.17 [−0.90; 0.56] 1.6%
Wiebenga 2015 −0.43 [−1.17; 0.30] 1.6%
Wiebenga 2015 −0.75 [−1.51; 0.00] 1.6%
Wiebenga 2015 −0.37 [−1.07; 0.32] 1.6%
Wiebenga 2015 0.35 [−0.34; 1.04] 1.6%
Wiebenga 2015 −0.52 [−1.26; 0.22] 1.6%
Wiebenga 2015 0.54 [−0.20; 1.28] 1.6%
Zhang 2024 −0.03 [−0.53; 0.47] 2.0%
Zhang 2024 −0.51 [−1.01; −0.00] 1.9%
Zhang 2024 −0.66 [−1.17; −0.15] 1.9%
Random effects model −0.19 [−0.35; −0.03] 100.0%
Heterogeneity: I^2 = 73.1%,
τ^2 = 0.2670, p < 0.0001
−3 −2 −1 0 1 2
B
Study
Glutamate+Glutamine
Cohen's d
95% CI
Weight
Arm 2021 0.05 [−0.58; 0.68] 3.2%
Arm 2021 −0.71 [−1.37; −0.06] 3.1%
Cawley 2015 −0.27 [−0.87; 0.33] 3.3%
Cawley 2015 −0.26 [−0.85; 0.34] 3.3%
Cawley 2015 −0.10 [−0.69; 0.50] 3.3%
Chard 2002 −0.24 [−0.78; 0.30] 3.5%
Chard 2002 −0.96 [−1.55; −0.37] 3.3%
Confort–Gouny 1993 −0.64 [−1.43; 0.15] 2.6%
Fernando 2004 0.38 [−0.11; 0.86] 3.8%
Imrich 2021 0.53 [−0.47; 1.52] 2.0%
Imrich 2021 0.25 [−0.74; 1.23] 2.0%
John 2024 −0.81 [−1.54; −0.09] 2.8%
John 2024 −0.10 [−0.79; 0.60] 2.9%
Kantorová 2017 0.04 [−0.49; 0.56] 3.6%
Kantorová 2022 −0.22 [−0.75; 0.31] 3.6%
Kantorová 2022 0.18 [−0.35; 0.71] 3.6%
Obert 2016 0.37 [−0.34; 1.09] 2.9%
Obert 2016 0.26 [−0.45; 0.97] 2.9%
Obert 2016 −0.12 [−0.88; 0.64] 2.7%
Obert 2016 0.26 [−0.50; 1.03] 2.7%
Polacek 2019 0.27 [−0.30; 0.84] 3.4%
Quadrelli 2019 0.63 [0.19; 1.06] 4.0%
Sastre–Garriga 2005 0.76 [0.23; 1.28] 3.6%
Sastre–Garriga 2005 0.18 [−0.33; 0.69] 3.7%
Solana & Martinez–Heras 2018 −0.32 [−0.76; 0.12] 4.0%
Solana & Martinez–Heras 2018 −0.14 [−0.58; 0.29] 4.0%
Tisell 2013 1.23 [0.50; 1.96] 2.8%
Tisell 2013 0.45 [−0.23; 1.12] 3.0%
Tisell 2013 1.13 [0.46; 1.79] 3.0%
Tisell 2013 0.41 [−0.21; 1.04] 3.2%
Xie 2025 −0.61 [−1.03; −0.20] 4.1%
Random effects model 0.05 [−0.13; 0.23] 100.0%
Heterogeneity: I^2 = 64.5%,
τ^2 = 0.1631, p < 0.0001
−3 −2 −1 0 1 2
C
MS – Control Effect Size
(Cohen's d ± 95% CI)
0.4
0.2
0.0
−0.2
−0.4
ns
Glutamate
Glutamate+
Glutamine